Anticancer Drugs

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OBJECTIVES:

- Describe the relevance of cell cycle kinetics to the mode of action and uses of different anticancer drugs
- Discuss the major classes of anticancer drugs : mechanisms of action , major toxicities and uses of the main drugs in each subclass
- Discuss the rationale underlying strategies of combination drug therapy

- **Cancer:** is a disease in which there is uncontrolled multiplication and spread within the body of abnormal forms of the body's own cells.
- **Differences between benign and malignant tumors:**
- Malignant tumors :Dedifferentiation, invasiveness, ability to metastasize (ability to spread to other sites through blood or lymphatics).
- Both types manifest uncontrolled proliferation.
- Differences from **normal** cell: uncontrolled proliferation, dedifferentiation, invasive and metastasize.

Approaches to treating established cancer

- Surgical excision of localized tumor (ideal).
- Irradiation.
- Chemotherapy (the role of each depends on the type of the tumor and its stage of development). Usually indicated for disseminated tumors and as supplement after surgical excision and irradiation.

- The **goal of therapy:** is to achieve cure by eradication of every cancer cell and if not possible; palliation is the goal.

Cell cycle specificity of drugs:

- Generally, ***rapidly dividing tumors*** (e.g. hematologic tumors) are *more sensitive to anticancer drugs*
- While non-proliferating cells **(Go phase**) are more resistant.

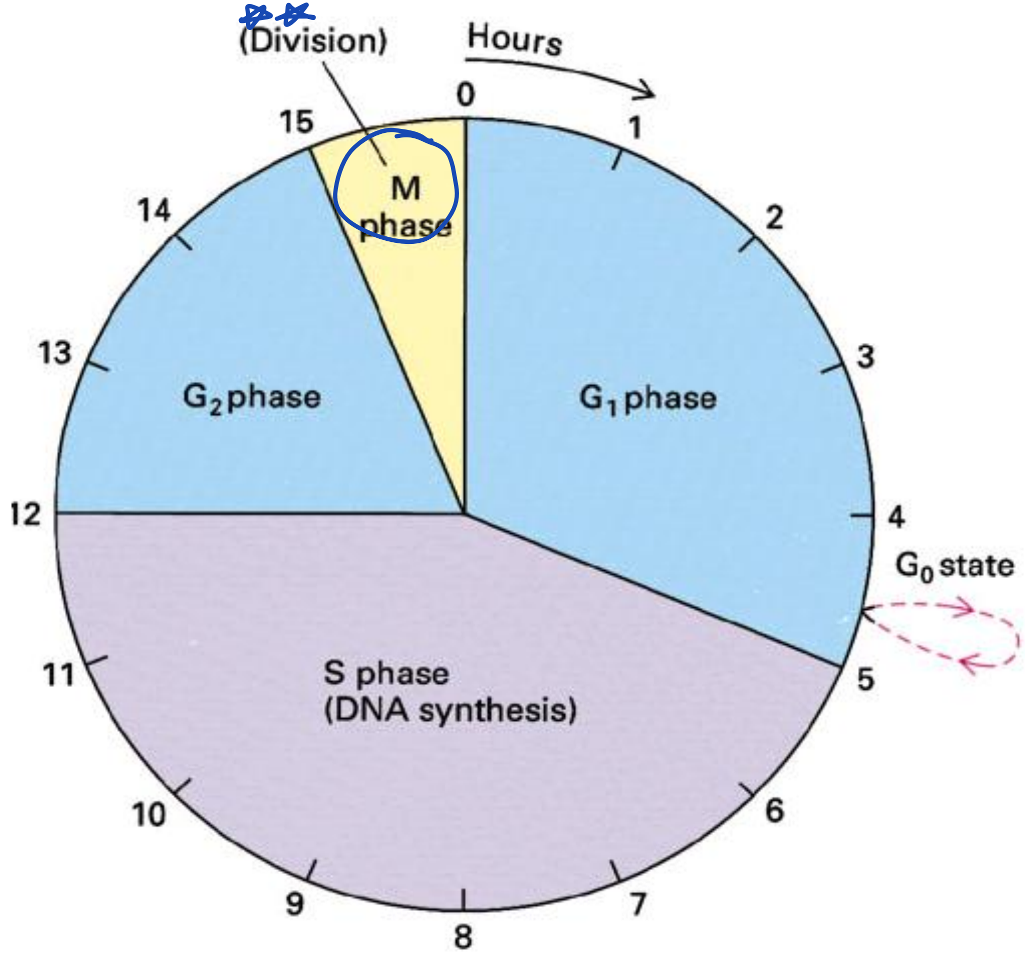
Cont.,

Mitotic Rapid Growth

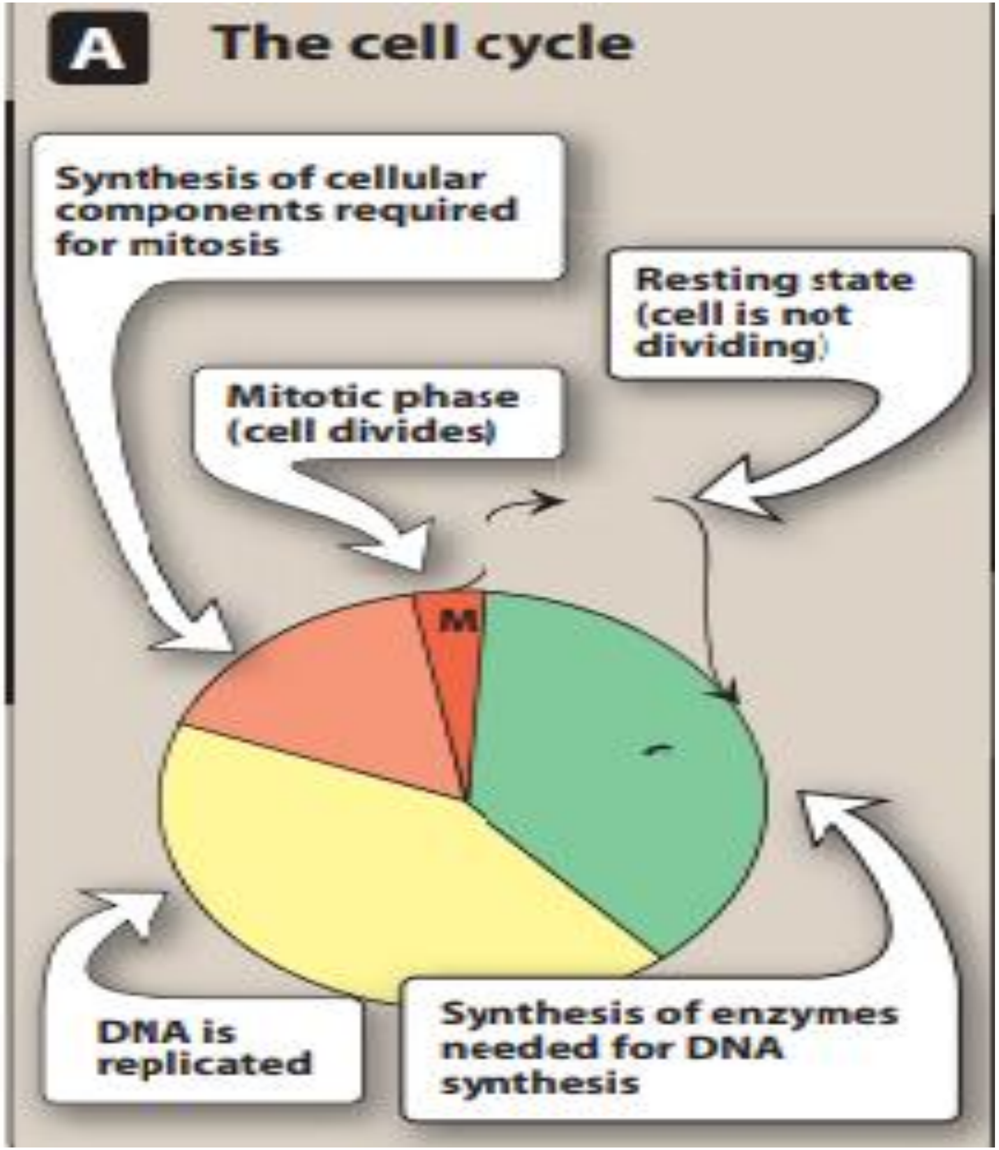
- Chemotherapeutic agents that are ***only effective against replicating cells*** are called ***cell cycle specific (CCS)***e.g. antimetabolites, bleomycin, vinca alkaloids and etoposide.

- while others are called ***cell cycle non specific*** **(CCNS)** and can affect ***low-growth and high-growth fraction malignancies***, e.g. alkylating agents, cisplatin, nitrosurias,

Cell cycle



(Division)
Hours
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
M phase
G_1 phase
G_2 phase
S phase
(DNA synthesis)
G_0 state


A
The cell cycle
Synthesis of cellular components required for mitosis
Resting state (cell is not dividing)
Mitotic phase (cell divides)
M
DNA is replicated
Synthesis of enzymes needed for DNA synthesis

B Cell-cycle specific drugs

Antimetabolites
Bleomycin peptide antibiotics
Vinca alkaloids
Etoposide

Effective for high-growth-fraction malignancies, such as hematologic cancers

C Cell-cycle non-specific drugs

Alkylating agents
Antibiotics
Cisplatin
Nitrosoureas

Effective for both low-growth-fraction malignancies, such as solid tumors, as well as high-growth-fraction malignancies

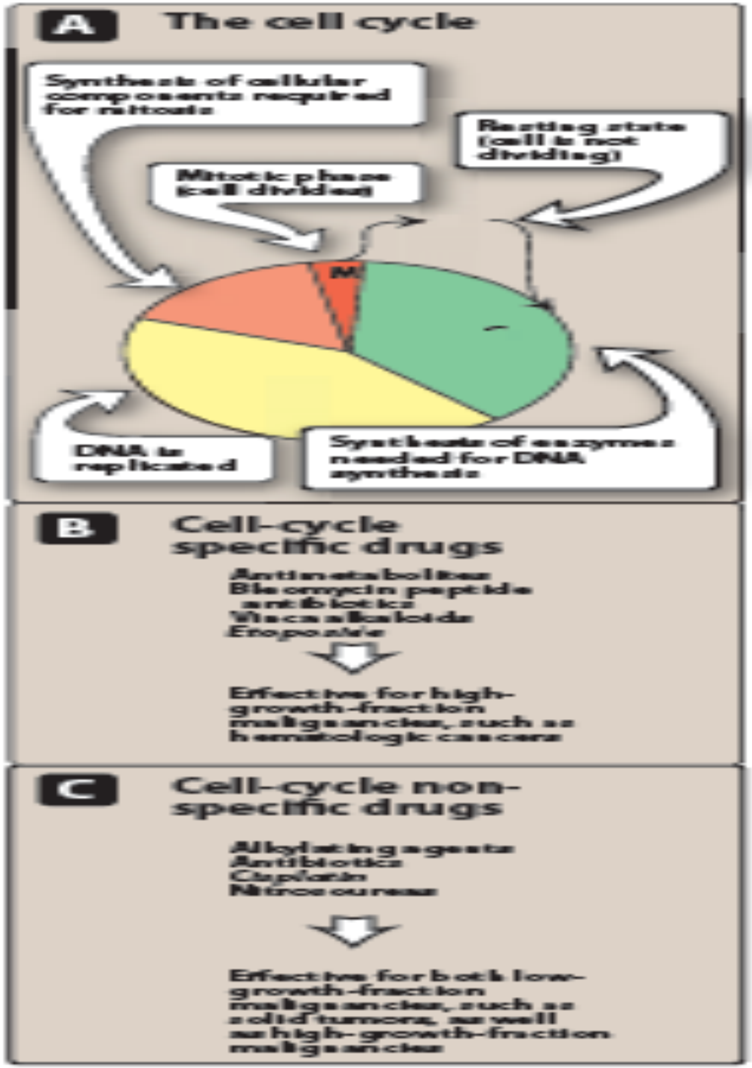


Figure 39.4
Effects of chemotherapeutic agents on the growth cycle of mammalian cells.

Treatment regimens:

- Usually according to body surface area and is best made specific to each patient.
- - Chemotherapeutic agents follows **first order kinetics** which means a constant **fraction** (percentage)of cells destroyed by a given dose.
- NB : Some tumor cells are present in **sites** that can't gain access to by certain chemotherapeutic agents as CNS, thus intrathecal administration or irradiation may be used.

Combination therapy

بس الناس الي عندها برست
كانسر ما تأخذ كومباينيشن
لانها أوردي تاخذ هرمون

*These combinations have **the advantages** of:*

- ***Maximal killing effect*** with tolerated toxicity.
- Affect ***broader range*** of cell lines.
- Delay or ***prevent resistance***.

- **Treatment protocols**: usually used intermittently and named by an acronym, e.g. POMP.

Problems associated with chemotherapy:

- 1-*Development of resistance*
- 2-***Multiple drug resistance (MDR).***
- 3-***Toxicity:*** cytotoxic drugs affect rapidly growing normal as well as cancer cells (e.g. cells of GIT mucosa, hair, bone marrow, buccal mucosa) →adverse effects.
- 4-***Treatment-induced tumors:*** As most of these agents are mutagens, tumors may arise many years (10 or more) after the cure of original tumor. This is specially a problem with alkylating agents.

Drugs used in cancer therapy:

*Q

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- **I-Antimetabolites:**

- synthetic analogues of normal metabolites, they interfere with purine or pyrimidine precursors either They are **CCS**.

- **1-Folate antagonists: Methotrexate (MTX)**
- **2-Purine analogues: 6- mecaptopurine**
- **3-Pyrimidine analogues: 5-Fluorouracil.**

1-Folate antagonists: Methotrexate (MTX)

- Folates are essential for **DNA synthesis and cell division**. In order to act as coenzyme, folic acid must be converted to FH4 (in 2 steps FH2→FH4) by ***dehydrofolate reductase (DHFR).***

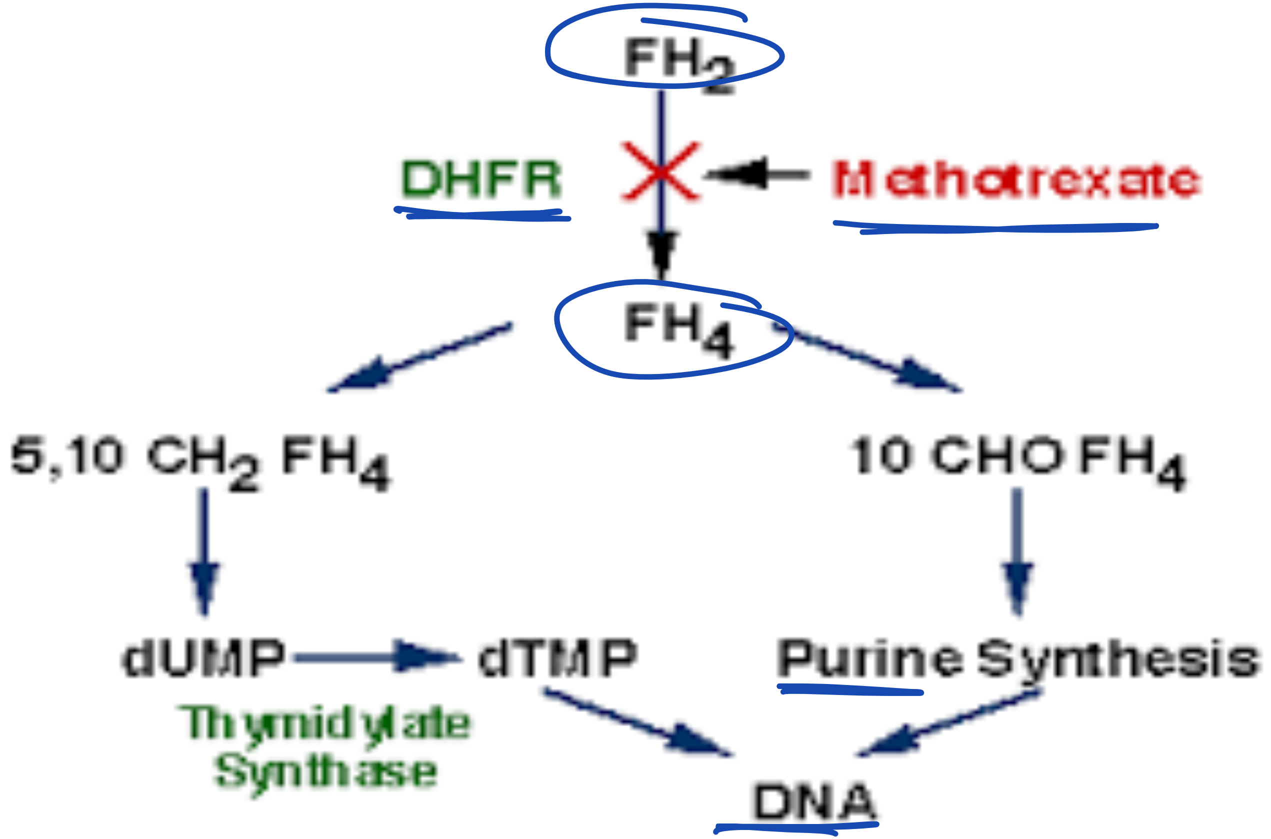
- **MTX** becomes *polyglutamated*. Both ***inhibit DHFR, inhibiting DNA, RNA and protein synthesis→cell death***.

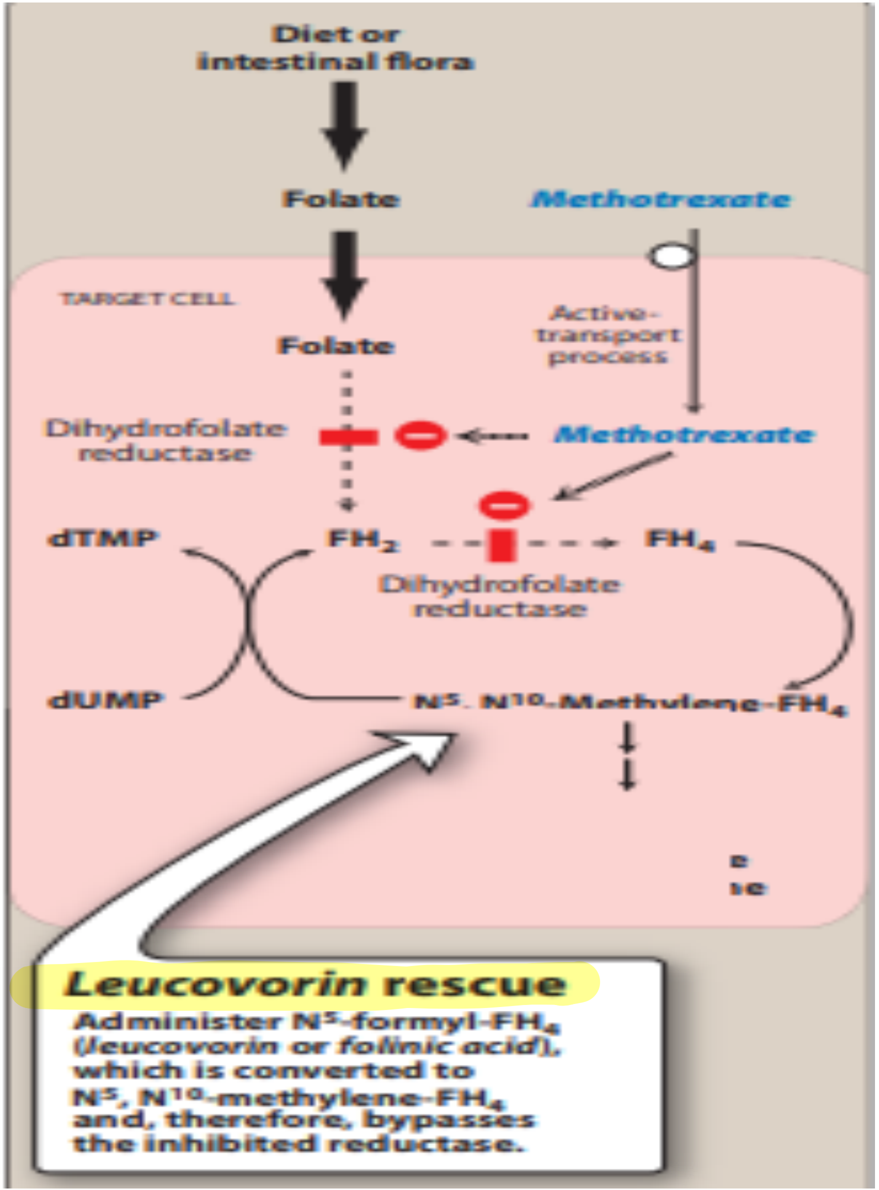
- Administration of **leucovorin** or folinic acid (***leucovorin rescue***, when fatal dose of MTX is used to maximize effect; the patient will die if not given) to bypass the blocked enzyme and replenish folate pool.

If the patient receives a high dose of MTX, leucovorin must be administered to rescue them

Sulfonamides: inhibit PABA \ Folic acid.
Trimethoprim: inhibits Folic acid \ Folinic acid (FH_4).
MTX (Methotrexate): inhibits FH2 \ FH_4 (by blocking DHFR enzyme)

Mechanism of action of methotrexate





Diet or intestinal flora
Folate
Methotrexate
TARGET CELL
Folate
Active-transport process
Dihydrofolate reductase
Methotrexate
dTMP
FH2
FH4
Dihydrofolate reductase
dUMP
N5, N10-Methylene-FH4
Leucovorin rescue
Administer N5-formyl-FH4 (leucovorin or folinic acid), which is converted to N5, N10-methylene-FH4 and, therefore, bypasses the inhibited reductase.

- **Adverse effects:**

- 1-Commonly seen with most anticancer drugs as ***nausea, vomiting, diarrhea, stomatitis, myelodepression and alopecia***. Some of these adverse effects are ***prevented or reversed*** by low dose leucovorin administration.
- 2-Renal damage: with high dose MTX , prevented by….
- 3-Hepatic toxicity by long term use.
- 4-Pulmonary toxicity: reversible.
- 5-Neurologic: with intrathecal administration(seizures, encephalopathy,…..

- **Therapeutic uses:**
- acute lymphocytic leukemia, breast cancer,…
- low dose is used in treatment of rheumatoid arthritis and psoriasis.

- **Resistance:**
- Resistance is associated with increased production of dihydrofolate reductase.

- **Contraindicated** in pregnancy (abortificient and mutagenic).

2-Purine analogues:

*** EXAM

Patients with gout taking allopurinol should have their 6-MP dose reduced, as both drugs interact via the same enzyme pathway (xanthine oxidase)

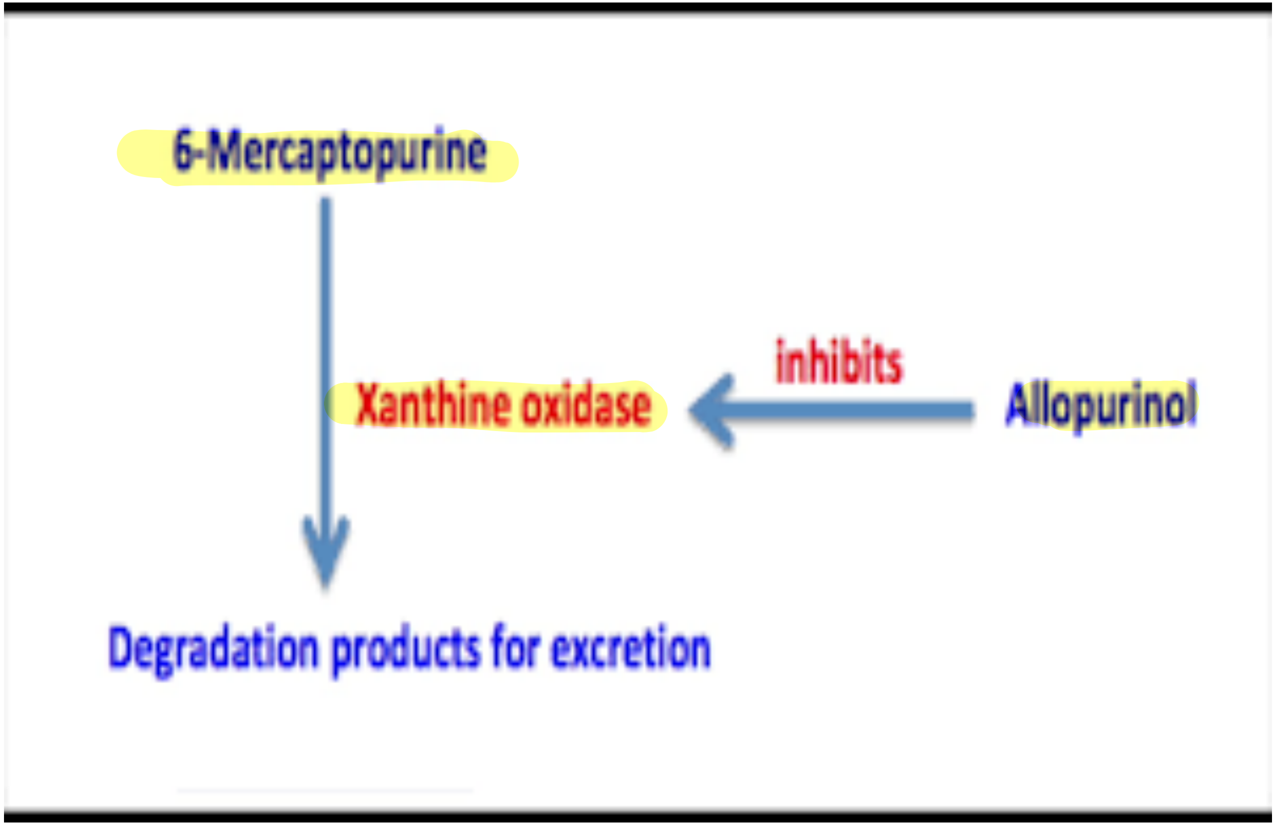
• 6-mercaptopurine (6-MP):

- It causes →non functional DNA and RNA .
- doesn't cross BBB.
- **Metabolized** to 6-MMP or thiouric acid by **xanthine oxidase** which is inhibited by **allopurinol** → accumulation of the drug and ↑ its toxicities So dose of 6-Mp ***should be decreased in patients taking allopurinol (by 25- 75%). Salicylates, NSAIDs, sulphonamides and sulphonylureas enhance the toxicity of MXT.***

Adverse effects:

- Myelodepression, GIT disturbances and hepatotoxocity.

Interaction of 6-mercaptopurine with Allopurinol



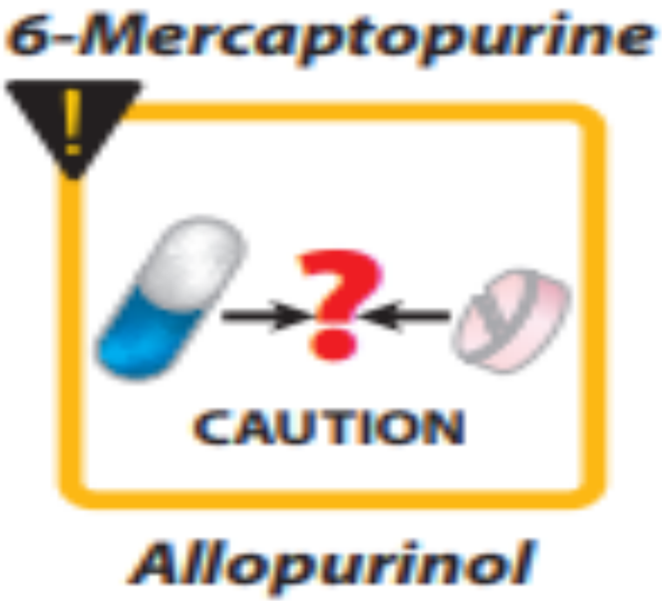


Figure 39.11
Potential drug interaction between *allopurinol* and *6-mercaptopurine.*

3-Pyrimidine analogues:

- **5-Fluorouracil:**

- **Mechanism of action:**

- Converted to FdUMP which competes with dUMP for **thymidylate synthase**.
- It forms a **complex** with the enzyme and its ***coenzyme leucovorin*** →↓ thymidine & DNA synthesis and cell death.

acts as a cofactor that enhances the efficacy

Adverse reactions:

- GIT Upset and severe oral and mucosal ulceration, alopecia, myelodepression.
- **Allopurinol** mouth wash may be given to decrease oral toxicity.
- ***Hand-foot syndrome*** is seen after extended infusions (i.e. erythematous desquamation of hands and soles).

- **Uses**: slowly growing solid tumors.
- ***N.B.*:**
- ***Leucovorin*** is administered concomitantly with **5-FU** as the reduced folate coenzyme ***is needed in the thymidylate synthase reaction***. Decreased coenzyme reduces the effectiveness of 5-FU.

II-Antibiotics:

نعرف الكلمتين المحدده بس

- They produce their effects mainly by direct action on DNA causing disruption of DNA function. They are **CCS**.

- **A -Bleomycin:**
- **Adverse effects :**
- **pulmonary** toxicity is dose-limiting ranging from cough to fatal fibrosis.
- Hypertrophic **skin** changes and hyperpigmentation of hands are prevalent, while myelodepresion is uncommon.

anti-HIV drug Nevirapine causes similar skin adverse effects

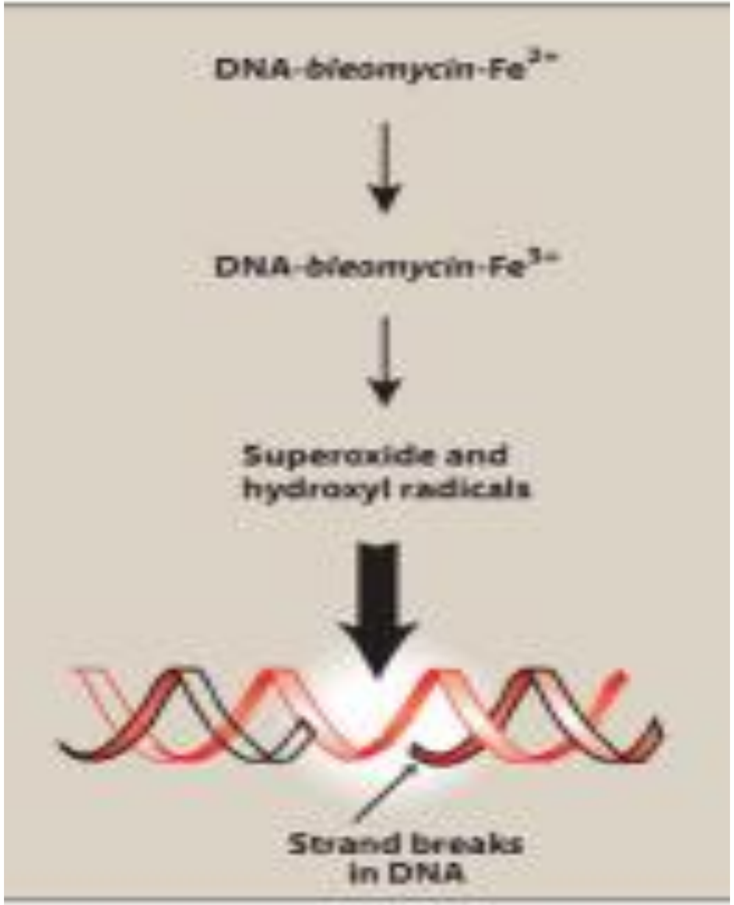


Figure 39.20
Bleomycin causes breaks in DNA by an oxidative process.

breaks in the DNA

- **B -Doxorubicin and daunorubicin:**

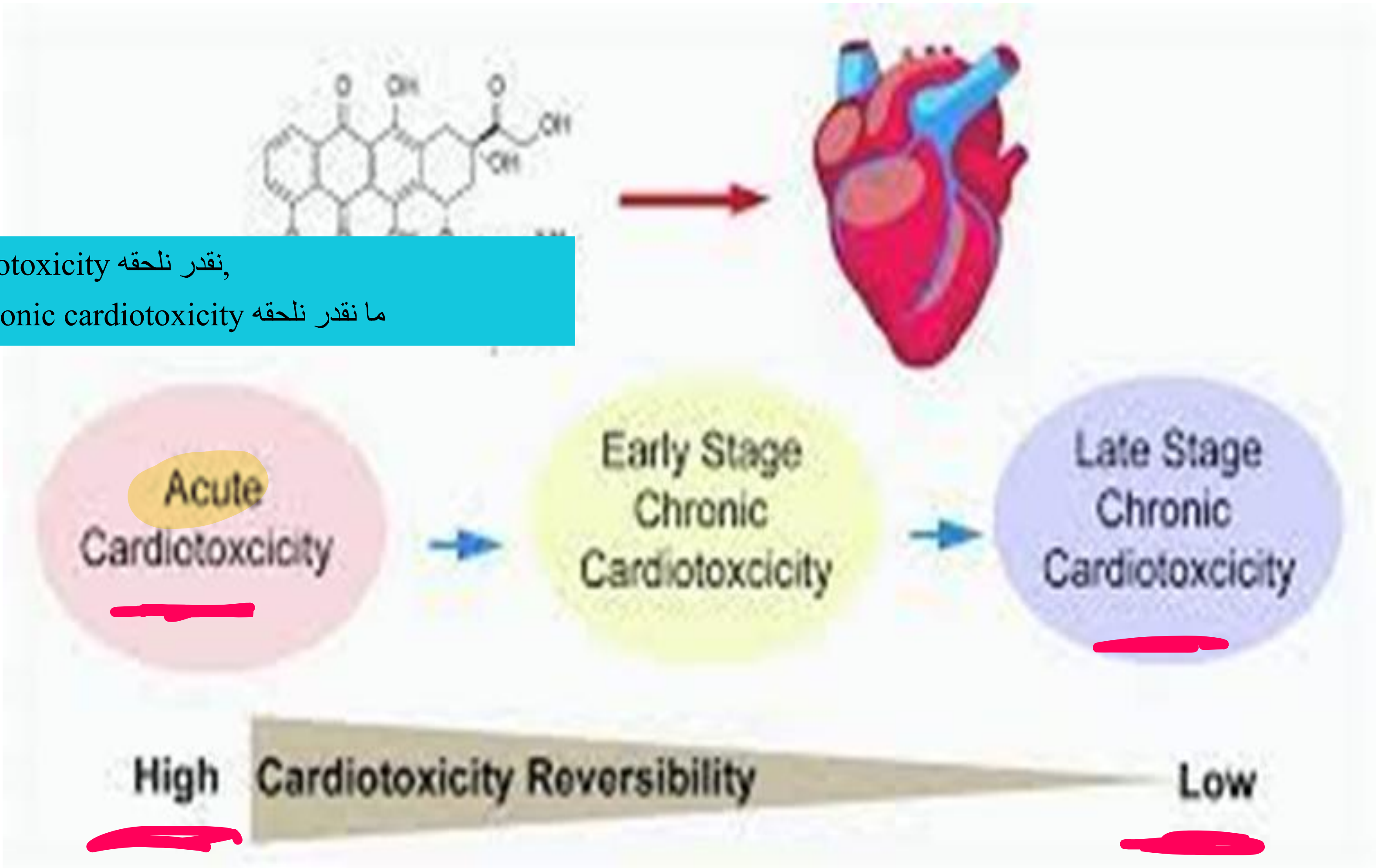
- Doxorubicin is used in combination with other agents for the treatment of sarcomas and a variety of carcinomas(*Wilms tumor*).

- **Mechanism of action:**

- ***Intercalation into DNA strands,*** also inhibits **topoisomerase II-** catalyzed breakage reunion→ ***irreparable DNA strand breaks*** → **free radicals** → single strand scission of DNA.

- The latter mechanism is **responsible** for the drug's **cardiac toxicity**.

Acute cardiotoxicity نقدر نلحقه,
whereas chronic cardiotoxicity ما نقدر نلحقه



Adverse effects

- Irreversible dose-related **cardiotoxicity.**
- Thorax irradiation and Trastuzumab (monoclonal AB) increase cardiac toxicity.
- ****Iron chelator *dexrazone*** may ***protect against cardiac*** toxicity, also *liposomal encapsulated formulation cause less cardiotoxicity*.

ذيه

antifungal drug Amphotericin B is used to decrease nephrotoxicity.

- The drug causes **reddish** discoloration of urine.
- **Common adverse effects of cytotoxic drugs are also encountered (GIT, myelodepression and alopecia).

Anticancer Drugs

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DR. DOAA BAKHASHWAIN

III-Alkylating agents

- They are **CCNS** *cellcyclenonspecific*
- ***All are mutagenic and carcinogenic, leading to a second malignancy as acute leukemia.***

• A -Mechlorethamine:

- **Mechanism of action:** it acts by transferring **alkyl groups** to N7 position of **guanine** in DNA resulting in either ***DNA strand breakage , cross linking of the two strands or intrastrand linking*** so that normal synthesis is prevented. Dividing cells are more sensitive to the action of the drug. Nitrogen mustard

The cell cannot replicate."

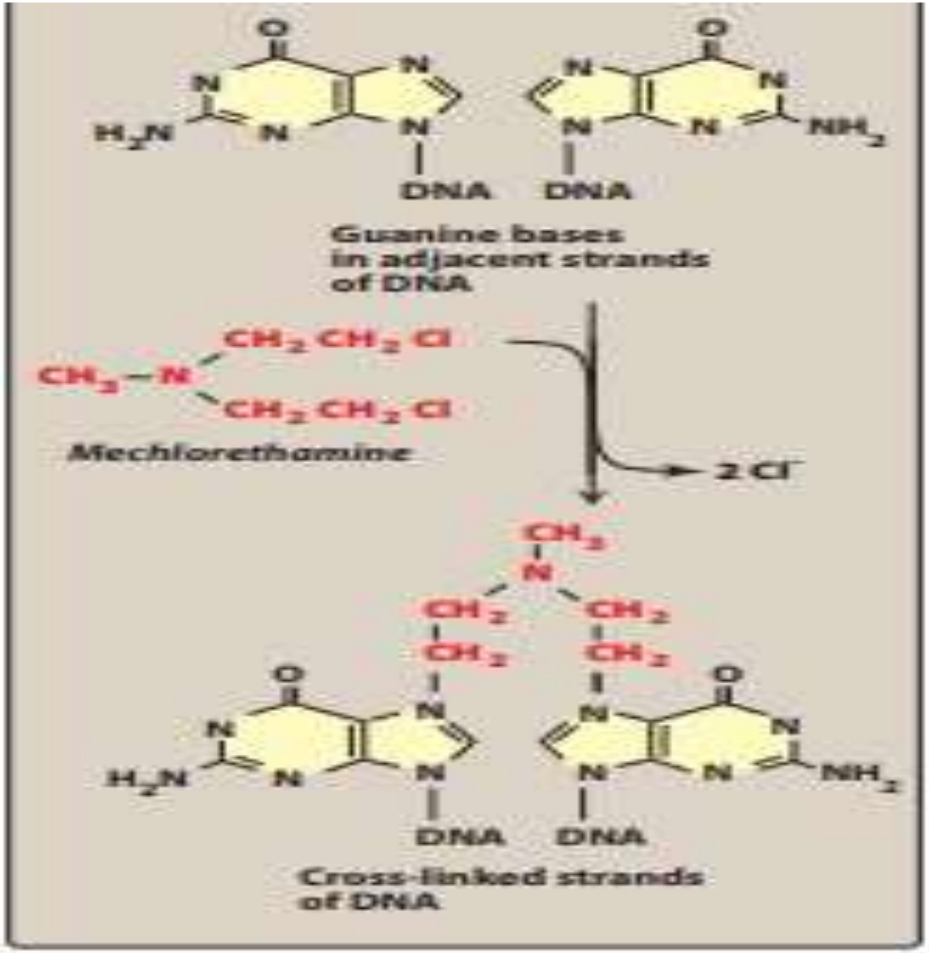
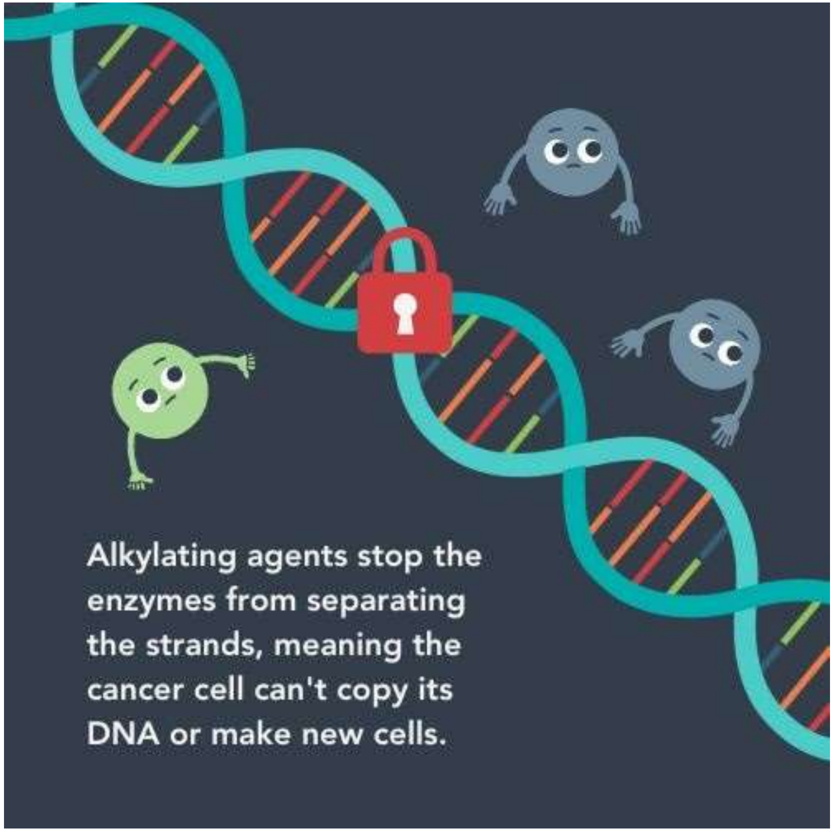


Figure 39.21
Alkylation of guanine bases in DNA is responsible for the cytotoxic effect of *mechlorethamine.*



Adverse effects:

- *Severe nausea and **vomiting** of central origin. This can be diminished by prior administration of **ondonsetron(**serotonin 5-HT_3 receptor antagonists.)_and dexamethasone.
- ***Bone marrow depression** is severe, latent viral infection (immunosuppressant) occurs.
- *If extravasation occurs, it must be treated with **sodium thiosulfite**.

B -Cyclophosphamide:

Orally administered or IV ☆☆

Adverse effects:

- nausea, vomiting, diarrhea and myelodepression.
- **Hemorrhagic cystitis** is caused by **acrolein** ***This toxicity can be diminished*** by increasing fluid intake and by **MESNA** (sodium-2-mercaptoethane sulphonate) which interact with acrolein forming a non-toxic compound. ☆☆ اهم شي
- **Amenorrhea, testicular atrophy and sterility.
- Veno-occlusive disease of liver, secondary malignancies.
- **Neurotoxicity

C -Nitrosureas:

- **Lomustine and carmustine** have the ability to pass BBB, thus used for treatment of ***brain tumors***.
- **Streptozotocin** is used in ***insulinomas*** since it is specifically toxic to β cells of islets of Langerhans.
- **Mechanism of action:** They function by cross linking through **alkylation** of DNA inhibiting its replication→ inhibition of RNA and protein synthesis.
- **Adverse effects:** delayed myelodepression and aplastic anemia on prolonged use, renal and pulmonary toxicity.
- ***Streptozotocin is diabetogenic***.

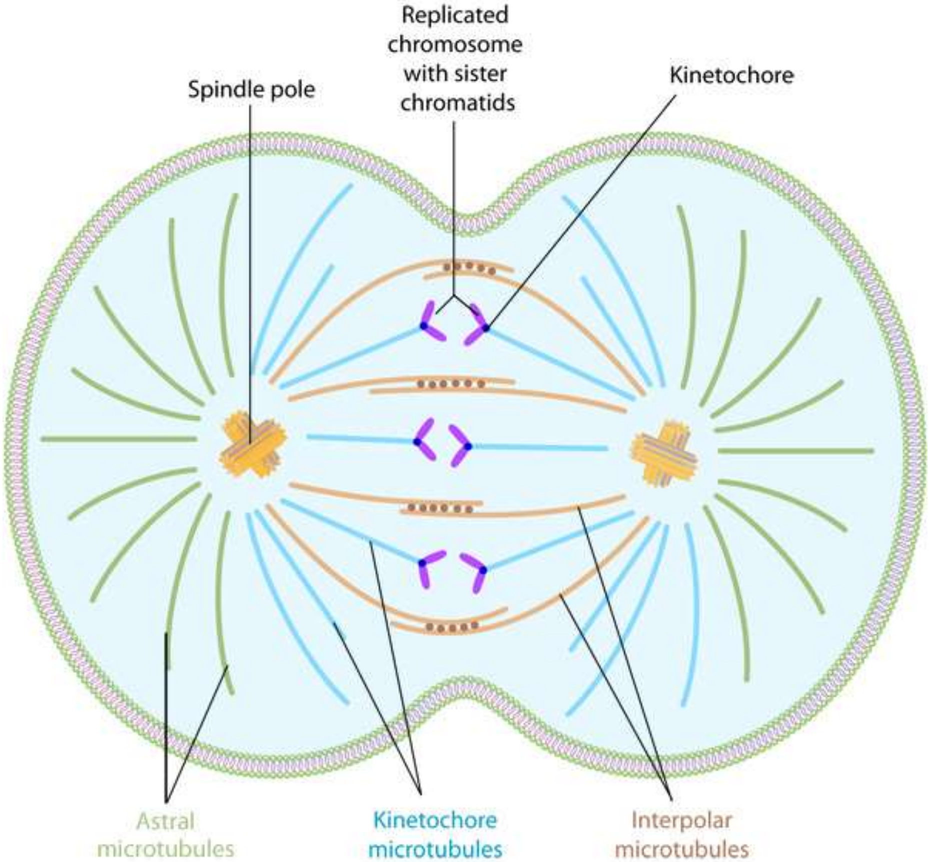
Nitrosureas are used to treat brain tumors due to their ability to cross the (BBB)
while streptozotocin is used for insulinomas because it is specifically toxic to beta cells, making it diabetogenic

IV-Microtubule inhibitors:

CCS

- Microtubules are important parts of cytoskeleton (needed for movements of cytoplasmic structures) and mitotic spindle (consisted of chromatides and microtubules composed of tubulin).
- The ***mitotic spindle is essential for equal portioning of DNA into two daughter cells during cell division***.
- These drugs act by affecting the equilibrium between polymerized and depolarized forms of microtubules.

Mitotic spindle



- **A-Vincristine (VX) and vinblastin (VBL):**

- Referred to as **vinca alkaloids**. VX is used in combination with other drugs for the treatment of leukemias (POMP), Wilms tumor, … While VBL is used for treatment of testicular carcinoma with bleomycin and cisplatin.

- **Mechanism of action:**

- They are **CCS** and phase specific.

Adverse effects:

- GIT, alopecia, if extravasation occurs→ phlebitis or cellulites.
- VBL is more myelodepressant than VX, while VX causes **neuropathy (areflexia : absence of reflexes) and muscle weakness** especially the quadriceps.
- **Constipation** is the most common symptom of autonomic neuropathy..

- **B -Paclitaxel and docitaxel:**
- **Adverse effects:**
- ***Neutropenia*** which is prevented by granulocyte colony-stimulating factor *(G-CSF)(**filgrastim**).*
- Serious **hypersensitivity** reaction necessitates pretreatment with dexamethasone, dyphenhydramine and H2 blocker.
- **Peripheral sensory neuropathy** and alopecia can occur.

V-Steroid hormones and their antagonists:

- Tumors that are steroid hormone sensitive may be either:

1-Hormone – **responsive**: in which tumor regresses *following giving* treatment with the specific hormone. This is only palliative except in case of lymphomas.

- 2-Hormone- **dependent (sensitive)** in which *removal* of hormonal stimulus causes tumor regression. Removal is achieved by surgery or drugs (tamoxifen in breast cancer).

A-Corticosteriods:

- **Prednisone**: it reduce the inflammation and help prevent an allergic reaction to cancer treatments. reduce your body's immune response, for example after a bone marrow transplant. help reduce sickness when having cancer treatment

- **Adverse effects:**

- hyperglycemia, hypertension, iatrogenic Cushing syndrome, cataract, glaucoma, osteoporosis, depression, euphoria, psychosis, peptic ulceration and delayed healing of wounds (immunosuppressive),…..

- **Uses:** in lymphomas and leukemias.

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- **B-sex hormones:**

ممكن أعطي معاهم أدوية سيولة

- **Estrogen and progestins** ; Estrogen can cause myocardial infarction , thromboembolic accidents , strokes, impotence ,… when used in treatment of cancer prostate.

C- sex hormones antagonists:

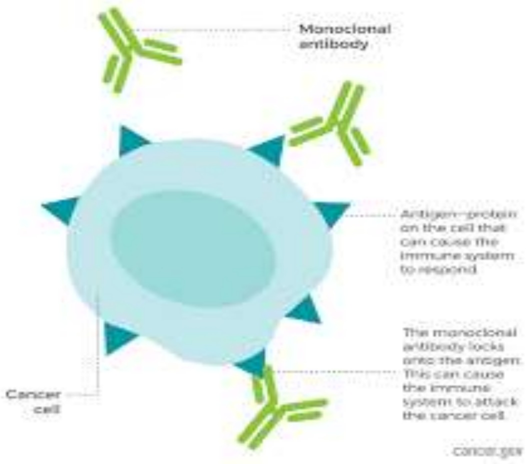
EXAM

- 1-**Tamoxifen:**- **CCNS**
- It is an ***anti estrogen*** drug with weak estrogenic activity.
- -A selective estrogen receptor modulator (SERM), also toremifene.
- -Used as 1st line treatment of estrogen receptor **positive breast cancer.**

VI-Monoclonal antibodies:

- ***Directed at specific targets and produce fewer adverse effects.***
- **-"Humanized"** ABs are now available by DNA recombinant technology by replacing part of mouse gene sequence with human genetic material.
- نعرف هذا بس

Humanized monoclonal antibodies target specific tumor types; however, their use is limited by high costs and مو متوفر في كل مكان



VII-Other chemotherapeutic agents:

- a-**Platinum coordination complexes:**
- **Cisplatin and Carboplatin**
- **Mechanism of action:** similar to alkylating agents.
- They are **CC non specific**. They can bind proteins and compounds containing (-SH) groups.

** MSQ

Combination therapy Enhancing efficacy while reducing resistance and side effects

Cont.,

- **<u>Adverse effects:</u>**

- severe **persistent vomiting** which is ameliorated by pretreatment with antiemetic drugs. **Nephrotoxicity** due to **Cisplatin's proximal tubular toxicity** which necessitates intensive hydration and diuresis**.** **Ototoxicity** and **neurotoxicity** also occurs and increased with concomitant administration of **aminoglycosides**

- **Alopecia due to outer hair cell toxicity.**

Care Capsule

B.Interferons (IFN):

- They are antineoplastic , antiviral and immunosuppressive .Classified into 3 types: α, β, γ.
- **Adverse effects:**
- **Depression** with alpha IFN. **Leukopenia** and thrombocytopenia are dose-limited toxicities.
- Fever and chills during 1st days of treatment.
- Elevation of **liver** enzymes may occur, also **nephrotoxicity** at high doses.
- **Loss** of weight, **hairs** and fatigue.
- Neuropathy and myopathy.